CS 599 / OSS RESEARCH TRACK

Verification Guidance in Reused Agent Skills

Understanding GitSkills and designing a feasible empirical study

Checkpoint 1 research report
Prepared October 23, 2026 (America/Phoenix)
Assignment due October 20, 2026 - late-submission version

Abstract

Agent skills package instructions and optional supporting files for software agents. This study asks whether explicit, actionable verification guidance is associated with reuse across repository owners, and whether verification steps that refer to local supporting files are backed by the recorded folder inventory. It treats a distinct skill text as the primary unit and separates content reuse from the properties of a particular repository copy.

For this checkpoint, the public GitSkills sample was downloaded and queried locally in read-only SQLite. The database contains 29,786 file occurrences, 13,000 distinct contents, 11,841 repositories, and 47,829 sibling entries. All 13,000 representative texts independently reproduce their Git blob hashes. The audit also found 16 enriched nonrepresentative copies, 1,755 truncated representative folder listings, and sharply unequal history coverage across location classes. These observations determine the inclusion rules and missing-data policy below.

The strict study frame contains 11,192 content groups. A deterministic, stratified 240-group coding frame has been generated, but manual annotation and research-question comparisons have not been performed. The proposed contribution is a validated measure of verification guidance and an analysis of its relationship to reuse and supporting material, rather than another inventory of skill counts.

Evidence boundary. Sample queries and quality checks are completed. Full-release analysis, human labels, and hypothesis tests are planned. The study repository is linked on page 8. No peer feedback was available; that requirement remains outstanding.

Reading guide

Dataset and resources: p. 2. Structure and observed example: p. 3. Questions and related work: p. 4. Method and validation: pp. 5-6. Quality evidence: p. 7. Work plan and peer-feedback status: p. 8. References: p. 9. Reproduction and SQL: pp. 10-12.

CHECKPOINT 1 / 01

Dataset overview and available resources

Dataset and release. GitSkills was created by Giuseppe Destefanis, Daniel Graziotin, Matteo Vaccargiu, and Marco Ortu to support empirical research on agent skills [2, 5]. The full archive is version 1.0.0, published August 10, 2026, DOI 10.5281/zenodo.21875637 [5]. This checkpoint queries its public sample, not the full release. Sources and sample were accessed October 23, 2026.

Exact analyzed snapshot. agent_skills_sample.db, 290,955,264 bytes, from the ZIP in sample repository revision `fff3df93beeee51a5444450891751f143e38da3e` (September 12, 2026). The ZIP was fetched through main and its Git blob hash was checked against that revision. Database and ZIP SHA-256 values appear on page 10. The sample is not assigned a separate release number.

Artifact / coverage	Full release: published [3-5]	Sample: observed
File occurrences	3,797,117	29,786
Distinct content hashes	1,877,981	13,000
Repositories / owners	282,200 / 195,841	11,841 / 10,786
Sibling entries / with text	7,264,865 / 3,497,752	47,829 / 22,278
File histories / mining runs	458,548 / 7	3,010 / 7

Resources and format. The preprint [2] explains the pipeline and structure; the sample README [3] is the field dictionary and supplies working SQL examples. The Hugging Face card [4] lists typed fields and a Python datasets loader. The full download is SQLite on Zenodo (44.4 GB) [5], with table-sharded Parquet on Hugging Face [4]. This study uses SQLite for transparent joins. No dedicated GitSkills notebook was verified; the verified examples are the README SQL and dataset-card loader. All URLs are in the references.

Collection and selection. Documentation describes read-only collection from GitHub code search, REST/GraphQL APIs, and raw content in July 2026. Searches were partitioned by file size to work around the 1,000-result limit. Content hashes group identical bytes; one preferred representative is enriched. Files lost before retrieval were excluded [3, 4]. The sample selects the 13,000 lowest content hashes and retains their occurrences and related records [3]. This approximates content-level sampling, not repository-level random sampling.

Dates and omissions. In the sample, discovery timestamps span July 9-20, 2026; first recorded commits reach back to April 14, 2025, while last commits extend through July 23, 2026. Collection dates describe retrieval, not creation. The data do not record agent invocation, execution success, user satisfaction, private repositories, or a complete edit history for every copy. Same bytes establish content identity, not authorship or copying direction.

License. The dataset card releases collected metadata and aggregation under CC BY 4.0; reproduced file text retains the origin repository license [4]. A missing repos.license is unknown permission, not a free-use designation. Reproduction materials here contain code, aggregates and identifiers; the raw dataset is downloaded separately.

CHECKPOINT 1 / 02

Data structure and relationships

The actual SQLite schema has four substantive tables plus the internal sqlite_sequence table. PRAGMA inspection and executed joins support the following model; the full DDL is included with the reproducibility materials.

Table and record unit	Identifiers and fields used
artifacts One repository-path occurrence	Primary key (repo_full_name, path). file_sha links identical bytes. filename, location_class, discovered_at identify discovery context; dedup_primary identifies the selected text. content, frontmatter_valid, description, body_chars and composition/history flags support eligibility and measurement.
repos One repository	Primary key full_name. Join artifacts.repo_full_name = repos.full_name. owner enables cross-owner counts; is_fork and metadata_fetched define known coverage; language, stars and license provide context.
artifact_siblings One listed folder entry	Primary key (repo_full_name, artifact_path, entry_name). Join its first two fields to artifacts.(repo_full_name, path). entry_type distinguishes file from directory; content and entry_size describe recorded material.
mining_runs One collector run	run_id identifies provenance; query, started_at, finished_at, discovered and note describe collection. No occurrence-level run foreign key is present.

Relationship model

```
repos.full_name  1 ---- * artifacts.repo_full_name
                          (repo_full_name, path) 1 ---- * siblings
                          file_sha * ---- 1 representative artifact
mining_runs: separate provenance log; no direct row-level join key
```

Annotated example from the downloaded sample

One representative is repository **obot-platform/obot**, path `.claude/skills/draft-release-blog/SKILL.md`, hash `00000f2a9d8359900f2075b536b435ca9473a5b9`. It has name draft-release-blog, body_chars = 13,056, sibling_count = 1, has_scripts = 1, and has_references = 0. Joining on repository plus artifact path returns md_to_html.py, a file of 1,158 bytes. This is an observed metadata example, not evidence that the script runs or that the skill contains effective verification.

Repeated content and important join rules

There are 3,378 multi-occurrence groups and 2,391 groups spanning multiple owners among all 13,000 groups. A repository can contain several paths with identical text, and an owner can control several repositories. Therefore COUNT(*) is an occurrence count, COUNT(DISTINCT repo_full_name) measures repository reach, and COUNT(DISTINCT owner) measures owner reach. None establishes independent adoption.

Filter text units by dedup_primary = 1, not content IS NOT NULL: 16 nonprimary rows also carry text. Join siblings to the chosen occurrence using the composite key, never file_sha alone. Representative folder contents must not be propagated to other copies. sibling_count agrees with file entries rather than all entries in complete inventories. The meanings of sibling content_fetched = 2 and missing skip reasons are not fully resolved; use actual text presence and explicit coverage flags.

CHECKPOINT 1 / 03

Research questions and related work

RQ1 - Verification guidance and cross-owner reuse

Among eligible distinct skill contents, how does the prevalence of actionable verification guidance differ between skills held by one occurrence, multiple occurrences under one owner, and multiple owners? This matters because reusable instructions may spread more broadly than their safeguards. A noncausal association can identify whether widely distributed skill texts deserve targeted review for clear checks, without equating popularity with correctness.

The unit is one file_sha content group. Required fields are representative content and body_chars; all eligible occurrences provide file_sha and repo_full_name; repos.owner supplies distinct-owner reach. Human labels measure verification specificity. The main comparison estimates the difference in actionable-guidance prevalence between the cross-owner and single-occurrence strata, with a same-owner stratum as a separate comparison.

RQ2 - Evidence behind verification steps

For eligible representative skills whose verification steps explicitly reference local supporting files, how often are those files present in the recorded complete folder inventory, and does support differ by verification specificity? This connects instruction quality to the material available to carry it out. The unit remains a content group represented by one repository-path folder; it is not every copy of that content.

Required fields are representative content, composition_fetched, composition_truncated, and sibling entry_name and entry_type. Manual coding isolates local file references associated with a check. The analysis compares the fraction of fully resolved local references for V2 versus V3 guidance and reports unresolved cases separately. No claim about execution success is possible from file presence alone.

What existing work establishes

GitSkills [2] establishes a corpus and a distinction between file occurrences and distinct content. Its data enable static reuse analysis but do not measure whether an agent follows a check. **Destefanis [7]** argues for evaluation-driven skill authoring and behavioral evaluation. That motivates a verification construct but does not establish its prevalence or reuse association.

Gao et al. [8] empirically study authoring, reuse and maintenance in registry and personal-use skills. They report substantial near-verbatim reuse and limited changes to behavioral contracts. Their broader content and modification analysis motivates examining explicit verification guidance with a reproducible, narrower rubric.

Seddik [9], submitted October 8, 2026, reconstructs dated copy networks, studies propagation and risk, and evaluates audit ranking. That work makes a generic copying-concentration study insufficiently distinctive here. This project instead examines verification instructions and their local supporting references; it does not reconstruct diffusion or infer source repositories.

Kalliamvakou et al. [10] show that GitHub repositories have heterogeneous purposes and that mined traces can misrepresent activity. Accordingly, repository counts are not treated as production use, and catalogs/templates will be recorded as contextual categories. The open format specification [6] requires metadata and allows arbitrary body instructions and optional supporting directories; valid front matter is therefore an eligibility proxy, not a quality judgment.

Expected addition. A small, auditable codebook; a human-labeled stratified sample; effect estimates for verification guidance and reuse; and a conservative check of local supporting-file presence. Novelty is scoped to this combination within the reviewed sources; no claim of exhaustive literature coverage or first-ever discovery is made.

CHECKPOINT 1 / 04

Proposed method

Design. A mixed quantitative and qualitative, cross-sectional study of the pinned GitSkills sample. The sample alone is the committed semester scope. A full-release replication is optional and depends on storage and time; it will not replace the pinned sample silently. The audit and feature extraction below are completed; semantic coding and inferential comparisons are proposed.

Population and selection

The primary frame keeps dedup_primary = 1, exact case-sensitive filename SKILL.md, frontmatter_valid = 1, and nonempty content, joined to a repository with metadata_fetched = 1 and is_fork = 0. There are 11,197 exact-name, parser-valid representatives; five have unknown fork status and are excluded, leaving **11,192 groups**. The parser flag does not certify every constraint of the current specification. A sensitivity analysis will relax filename/header restrictions and show exclusions explicitly.

Reuse is computed over occurrences with the exact basename and known nonfork repository status, grouped by file_sha. All three strata refer to this restricted occurrence set: **8,265 single-occurrence**, **693 same-owner multiple-occurrence**, and **2,234 cross-owner** groups. The representative need not be the first or original author. Unknown repository status is not recoded as nonfork.

Select 80 groups per stratum (240 total), ordered within stratum by SHA-256 of cs599-label-v1: followed by file_sha. This selection was executed and the blank coding form is included. Hide strata, reach counts and stars during coding. The design overrepresents reused groups to enable comparison; frame-level prevalence uses weights N_h/80. The unequal stratum counts do not change within-stratum estimates.

Operational definitions

Concept	Proposed measurement
Verification specificity	V0: no directed output check. V1: generic request to test/review. V2: concrete check action OR observable acceptance criterion. V3: both, explicitly connected. The primary actionable label is V2/V3. Uncertain cases remain unresolved.
Reuse / context	Cross-owner = more than one distinct owner in eligible occurrences. Record occurrence, repository and owner reach separately. Manually note software-task versus other purpose and catalog/template context. Presence does not imply invocation.
Supporting material	For a local file explicitly used by a verification step: normalize the relative path and match sibling entry_name where entry_type = file, in that representative folder only. Distinguish present, absent in complete inventory, and unknown/ambiguous.

Question-to-analysis mapping

RQ1: file_sha unit + human guidance label + owner/occurrence reach. Report stratum proportions, their cross-owner versus single-occurrence risk difference, and an odds ratio. Generate 95% intervals by 10,000 within-stratum bootstrap resamples (fixed seed 599); use weighted summaries only when combining strata. Treat the third stratum as a secondary comparison.

RQ2: representative folder unit + human-linked local paths + inventory coverage. Among complete folders with eligible references, compare per-skill resolved-reference fraction for V2 versus V3 using a bootstrap difference in means, resampling skills rather than individual references. Report denominators, uncertainty and examples; use descriptive results when either group has fewer than 20 skills. No manual labels or effect estimates exist yet.

CHECKPOINT 1 / 05

Measurement validation and interpretation

Human coding and checks

Pilot 30 of the 240 selected groups, refine ambiguous rules, freeze codebook v1, and recode those 30 before final analysis. Read the whole skill, distinguishing directed checks from background prose and illustrative user tasks. Record evidence line numbers and reasons for each label. Invented codebook examples: "review the result" is V1; "run pytest and require all tests to pass" is V3. These examples are not reported dataset findings.

Blindly recode 60 items after at least seven days: the 30 pilot items plus 30 additional deterministic selections. Report raw agreement, weighted Cohen kappa for V0-V3, and uncertainty. This is **intra-rater stability**, not independent inter-rater reliability. If weighted kappa is below 0.60, clarify rules and recode affected cases; if stability remains weak, collapse to actionable/nonactionable and limit interpretation. An independent second reader is desirable but is not available or claimed.

For RQ2, independently recheck every coded absent-file case against the composite-key inventory join and the literal reference context. Exclude generated outputs, placeholders, absolute paths, URLs, and paths clearly belonging to the user's external project. Mark ambiguous bases and symlink references unknown; never classify a missing text body as a missing file. An existing entry establishes recorded presence only, not that it is executable, safe, current, or usable.

Automation policy. No LLM classifier is used for reported findings. SQL and deterministic hashing handle structure and selection. If an LLM is added later, reserve an untouched human-labeled holdout and publish the model/version, prompt, confusion matrix, per-class precision/recall and abstentions. Agreement between models will not substitute for human validation. The prepared coding frame contains no completed semantic labels.

Analysis safeguards

As an exploratory adjustment for RQ1, fit cross-owner versus single-occurrence logistic regression using actionable guidance, log(1 + body_chars), and a software-task indicator only if data support at least ten observations of the less frequent outcome per coefficient and no separation. Otherwise keep unadjusted contrasts. Do not search many model specifications for significance. Inspect a sensitivity analysis excluding catalogs/templates and another retaining one representative per owner; explain that representative ownership does not capture every dependency across shared owners.

For unresolved labels, report their count and repeat prevalence contrasts with unresolved cases assigned to each extreme. For incomplete folder inventories, give both complete-case estimates and worst-case presence bounds over eligible references. Do not impute absent support from a truncated listing. These are planned sensitivity checks, not results.

Threats, scope and ethics

The snapshot cannot establish that verification guidance causes reuse or that it improves outputs. Shared templates, task purpose, text length and repository selection may confound associations. Exact hashes miss edited and near-duplicate skills. Hash-based content sampling is not a probability sample of repositories, owners or all GitHub activity. Header validity excludes some genuine skills and may include unrelated instructions. File history is path-specific and heavily selected, so maintenance is outside the primary RQs.

Analyze instructions as text; do not execute downloaded commands or attempt to reverse anonymized author codes. Report aggregates and short, license-aware examples. A missing check or support file is not evidence of malice. This report was prepared with AI assistance for source retrieval, analysis execution and writing; semantic research labels remain to be completed and reviewed by the student.

CHECKPOINT 1 / 06

Initial data-quality checks: observed evidence

All figures on this page are actual results from the pinned sample, produced by checks.sql and run_checks.py. They describe this sample only. The companion results file includes denominators, runtime versions and the complete seven-row collection log.

Check and denominator	Observed result	Handling / interpretation
Integrity and content identity	SQLite integrity_check = ok. 0/13,000 representative blob-hash mismatches. 0 missing hashes.	Trust byte identity for sampled text; this does not validate meaning or execution.
Relationships and representatives	0 orphan artifact-to-repo joins; 0 orphan sibling-to-artifact joins; all 13,000 groups have exactly one primary.	Use logical joins on observed keys. SQL schema declares primary keys, but no foreign-key constraints.
Unexpected enrichment	16/16,786 nonprimary rows have content and composition; 13 sibling rows join to nonprimary occurrences.	Always select dedup_primary = 1 for text units and their folders. Cause is unresolved; do not count extra text rows as new skills.
Filename/header population	378/29,786 names differ from SKILL.md; 1,709/13,000 representatives have frontmatter_valid = 0.	Use exact basename and parser-valid primary content for the main frame; relax restrictions in sensitivity checks.
Folder coverage	10/13,000 representatives lack composition; 1,755/13,000 listings are truncated (13.50%).	Missing/truncated folders cannot establish absence. 10,079/11,192 strict units have fetched, untruncated composition.
Sibling count semantics	2,915 mismatches if counting files + directories; 0 if counting only files, for fetched/untruncated representatives.	Count entry_type = file to compare with sibling_count. The apparent discrepancy is explained by directory rows.
Sibling text availability	22,278/47,829 entries contain text; 39,046 entries are files and 8,783 are directories. All skipped_reason values are NULL.	71 file rows have content_fetched = 1 but NULL content. Use content IS NOT NULL; missing text reasons remain unverified.
Repository missingness	728/11,841 missing language; 6,331 missing license; 37 unknown fork flags and star counts; 0 missing owners.	Keep unknown categories; exclude five otherwise-eligible primary units with unknown fork status. No license inference from NULL.
Dates and provenance	0 parse failures in present discovery/first/last dates; 0 reversed first/last dates; 2 histories precede Oct. 2025. Four runs have retroactively closed zero counts.	Use julianday rather than lexical timestamp ordering. Do not sum run counts as a corpus census or use file dates as format-adoption dates.

Uneven history coverage. Histories exist for 1,755/1,762 canonical representatives (99.60%), 813/7,416 skills-dir representatives (10.96%), and 442/3,822 other representatives (11.56%). Treating the 3,010 available histories as a random content sample would bias maintenance comparisons. The main study therefore does not require commit-history fields.

Risks still untested: near duplicates, symlink prevalence, correctness of front-matter parsing, true reference resolvability, label agreement, and success of any verification action. No prevalence or causal claims about these are reported.

CHECKPOINT 1 / 07

Work plan, responsibilities and peer feedback

Working arrangement. The plan assumes individual work because no teammate was identified in the recovered requirements. The student owns data provenance, code, all manual coding, analysis, interpretation, writing and the oral explanation. If a second member is formally added, divide coding and analysis while both review the full pipeline and design; do not retroactively claim two-person validation.

Date / milestone	Deliverable and responsibility	Dependency / contingency
Oct. 23-26	Finalize checkpoint package; review SQL and report; verify instructor repository access; request feedback from another group. Student leads all tasks.	Repository access and peer availability. Submit the honest checkpoint version under the available late-submission process.
Oct. 27-Nov. 2	Pilot 30 labels, freeze codebook, code remaining 210, validate reference matching, prepare descriptive summaries.	About 20-25 hours for coding at 5-6 minutes per item. If over budget, freeze a reduced balanced sample of 40 per stratum and report reduced precision.
Nov. 3 progress report	Show audit, pinned sample, codebook, annotation progress, initial stratum estimates and explicit unknowns. Demonstrate a reproducible query.	If coding is incomplete, present completed-label denominators and a completion date; do not label planned work as done.
Nov. 4-17	Blind recode after seven days; resolve disagreements; run planned contrasts and sensitivity checks. Optional full-release replication only after sample analysis.	Second reader and 44.4 GB download are optional. Keep sample-based claims if storage or access blocks replication.
Nov. 18-Dec. 2	Complete interpretation, charts, limitations and reproducibility audit; revise after available peer feedback; rehearse.	If reference resolution proves ambiguous, narrow RQ2 to hand-verified complete folders and report unknown cases.
Dec. 3 presentation	Present questions, observed data-quality findings, final sample results, uncertainty and limitations; explain units and joins.	Student must be able to rerun SQL and explain label decisions.
Dec. 4-9 / Dec. 9 paper	Incorporate presentation feedback, verify claims against outputs, finalize paper and archive code/manifest/results.	Confirm instructor access to the private repository and preserve the final analysis version.

Peer feedback and response to review

No feedback from another group was supplied or recovered. Consequently, this report contains no invented reviewer, quotation, review date or revision attributed to a peer. A response-to-peer-review record cannot yet be completed. Once feedback is obtained, record the reviewer/group, date, exact or faithful feedback, resulting revision and page, and a reason for any suggestion not adopted. The literature-informed scope changes and audit-driven safeguards in this version are preparation decisions, not peer feedback.

Study repository. https://github.com/paulrevanthpersonal-lab/cs599-gitskills-research. The private repository contains the read-only analysis runner, executed SQL, observed results, snapshot manifest, proposed codebook, blank coding frame and checkpoint report. The instructor needs repository access to inspect it. The raw dataset is downloaded separately from the authors' public source [3]. A local reproducibility ZIP also accompanies the report.

In-class check-in. The PDF cannot complete the oral component. Demonstrate the sample row count, representative invariant, and sibling-file count check (pp. 11-12), explain why content and occurrence units differ, and describe the planned label protocol. Do not imply that the check-in has occurred.

CHECKPOINT 1 / 08

References and verified source URLs

All online sources were accessed October 23, 2026. Preprints are cited as preprints; future conference publication is not treated as completed publication. The MSR page is conference context, while the CS 599 rubric governs this report.

[1] MSR 2027 organizers. "Mining Challenge." Official call. The page identifies GitSkills and SpecMine, links the releases, and lists MSR in Dublin, April 26-27, 2027.
https://conf.researchr.org/track/msr-2027/msr-2027-mining-challenge

[2] G. Destefanis, D. Graziotin, M. Vaccargiu, and M. Ortu. "GitSkills: A Dataset of Agent Skills on GitHub." arXiv:2608.10906v3, September 11, 2026.
https://arxiv.org/html/2608.10906v3

[3] GitSkills authors. gitskills-sample repository and README: field dictionary, collection notes, sample definition and example SQL. Pinned revision fff3df93beeee51a5444450891751f143e38da3e.
https://github.com/giuseppedestefanis/gitskills-sample/tree/fff3df93beeee51a5444450891751f143e38da3e

[4] M. Vaccargiu et al. GitSkills dataset card and Parquet mirror. Includes typed schema, Python loading example and license conditions.
https://huggingface.co/datasets/mvaccargiu/gitskills/blob/main/README.md

[5] G. Destefanis, D. Graziotin, M. Vaccargiu, and M. Ortu. GitSkills dataset, version 1.0.0, August 10, 2026. Zenodo. DOI: 10.5281/zenodo.21875637.
https://zenodo.org/records/21875637

[6] Agent Skills. "Specification." Open format specification for SKILL.md and optional supporting directories. Living documentation; accessed version, not assumed to equal July 2026 tooling.
https://agentskills.io/specification

[7] G. Destefanis. "Authoring Agent Skills: A Software-Engineering Approach." arXiv:2607.25032v1, July 27, 2026.
https://arxiv.org/abs/2607.25032v1

[8] H. Gao, J. L. Lulla, H. Y. Lin, S. Baltes, C. Treude, and M. Zahedi. "From Registry to Repository: How AI Agent Skills Are Written, Adapted, and Maintained." arXiv:2607.00911v2, July 6, 2026.
https://arxiv.org/abs/2607.00911v2

[9] F. Seddik. "Skill Constellations: Tracing the Supply Chain of Agent Skills on GitHub." arXiv:2610.11169v1, October 8, 2026.
https://arxiv.org/abs/2610.11169v1

[10] E. Kalliamvakou, G. Gousios, K. Blincoe, L. Singer, D. M. German, and D. Damian. "The Promises and Perils of Mining GitHub." MSR 2014, pp. 92-101. DOI: 10.1145/2597073.2597074. Author-hosted text inspected.
https://gousios.org/pub/promises-perils-github.pdf

Source reconciliation

The sample README still describes full archival as future work. The official call, Zenodo record and Hugging Face card provide the full release links [1, 4, 5]; therefore the report uses those for current availability. The README's approximate 277 MB/41 GB sizes are not substituted for observed byte counts or Zenodo's displayed 44.4 GB. The full-release counts remain published figures because that database was not downloaded.

CHECKPOINT 1 / 09

Reproducibility and execution record

Completed environment. Python 3.12.14 with SQLite 3.53.1, using only the Python standard library for data analysis. The SQLite connection uses mode=ro. run_checks.py executes the named SQL blocks, writes JSON results and the actual schema, independently recomputes representative Git blob hashes, extracts the strict study frame, and creates a blank 240-row coding form. It never executes dataset contents.

Download and identity

Sample source: authors' sample repository. Download agent_skills_sample.zip at the pinned revision in reference [3], then unzip it. The archive contains one database. The original download was verified against the revision's Git tree, which records ZIP blob SHA-1 b2c5b19033285d2361a20df94d36e3eb922b6a1a.

```
ZIP SHA-256 (86,721,947 bytes):
28a83ce750abde8fb904bc23191cb877a3645a1600eb326069fb2c18e7246fe8

Database SHA-256 (290,955,264 bytes):
683888c923a51952fb66f83f21351ed5bf1127ddddf380e517592ae27c367658
```

Portable rerun commands

```
unzip agent_skills_sample.zip
python3 run_checks.py agent_skills_sample.db --out rerun-results

# Or demonstrate one initial query with the SQLite CLI:
sqlite3 -readonly agent_skills_sample.db \
  "SELECT COUNT(*) FROM artifacts;"
# Observed output: 29786
```

The Python runner needs no SQLite command-line installation. It accepts any database path, exports results locally, and records the database hash and SQLite runtime version. A new release or modified database requires a new manifest and rerun; results should not be compared under an unchanged snapshot label.

Companion archive contents

File	Purpose and current status
checks.sql / study.sql	Full executed quality queries and content-group feature extraction. Appendix SQL below is a readable subset.
run_checks.py / manifest.json	Read-only runner; source URLs, snapshot pin, source-evidence hashes and database identity.
results/observed_results.json / schema.sql	Observed query outputs and actual database DDL. These are completed evidence.
results/study_summary.json	Observed 11,192 eligible groups, stratum counts and 10,079 complete folder inventories.
codebook.txt / label_sample_uncompleted.csv	Proposed protocol and reproducibly selected 240-group form. Semantic labels are blank and no coding results are claimed.

Hash audit detail. For each primary content string, encode UTF-8, prepend the Git header bytes blob + space + byte-length + NUL, and calculate SHA-1. Compare to file_sha. All 13,000 matches are independent of the dataset's content_sha_ok flags. A hash match verifies retained bytes; it does not validate source licensing, semantics, or folder content.

CHECKPOINT 1 / 10

Appendix A - executed quality-check SQL

Queries run against agent_skills_sample.db. Full queries and JSON outputs are in the companion archive. Results below are observed, not illustrative.

A1. Row counts and one representative per content

```
SELECT COUNT(*) AS occurrences, COUNT(DISTINCT file_sha) AS contents
FROM artifacts;
-- 29786 | 13000

SELECT COUNT(*) AS bad_groups FROM (
  SELECT file_sha FROM artifacts GROUP BY file_sha
  HAVING SUM(dedup_primary = 1) <> 1
);
-- 0
```

A2. Logical relationship integrity

```
SELECT COUNT(*) FROM artifacts a LEFT JOIN repos r
  ON r.full_name = a.repo_full_name
WHERE r.full_name IS NULL;
-- 0 orphan artifacts

SELECT COUNT(*) FROM artifact_siblings s LEFT JOIN artifacts a
  ON a.repo_full_name = s.repo_full_name
 AND a.path = s.artifact_path
WHERE a.path IS NULL;
-- 0 orphan sibling entries
```

A3. Test text-selection assumptions and missing coverage

```
SELECT COUNT(*) FROM artifacts
WHERE dedup_primary = 0 AND content IS NOT NULL;
-- 16: content IS NOT NULL alone is not a representative selector

SELECT COUNT(*) AS representatives,
  SUM(frontmatter_valid = 0) AS invalid_header,
  SUM(composition_fetched = 0) AS missing_folder,
  SUM(composition_truncated = 1) AS truncated_folder
FROM artifacts WHERE dedup_primary = 1;
-- 13000 | 1709 | 10 | 1755

SELECT location_class, COUNT(*) AS representatives,
  SUM(history_fetched = 1) AS histories
FROM artifacts WHERE dedup_primary = 1
GROUP BY location_class ORDER BY location_class;
-- canonical  1762 1755
-- other      3822  442
-- skills-dir 7416  813
```

A4. Explain sibling-count semantics

```
SELECT COUNT(*) AS mismatches FROM artifacts a LEFT JOIN (
  SELECT repo_full_name, artifact_path,
    SUM(entry_type = 'file') AS n
  FROM artifact_siblings GROUP BY 1, 2
) s ON s.repo_full_name = a.repo_full_name
   AND s.artifact_path = a.path
WHERE a.dedup_primary = 1 AND a.composition_fetched = 1
  AND a.composition_truncated = 0
  AND COALESCE(a.sibling_count, 0) <> COALESCE(s.n, 0);
-- 0; replacing SUM(entry_type = 'file') by COUNT(*) yields 2915
```

CHECKPOINT 1 / 11

Appendix B - executed study-frame SQL

This extraction creates analysis units and coverage flags. It does not label verification guidance, infer copying direction, or test the research questions. The full version also exports context and supporting-file counts.

```
WITH reuse AS (
  SELECT a.file_sha, COUNT(*) AS occurrences,
    COUNT(DISTINCT a.repo_full_name) AS repositories,
    COUNT(DISTINCT r.owner) AS owners
  FROM artifacts a JOIN repos r
    ON r.full_name = a.repo_full_name
  WHERE a.filename = 'SKILL.md' COLLATE BINARY
    AND r.metadata_fetched = 1 AND r.is_fork = 0
  GROUP BY a.file_sha
), frame AS (
  SELECT p.file_sha, p.repo_full_name, p.path, p.body_chars,
    p.composition_fetched, p.composition_truncated,
    u.occurrences, u.repositories, u.owners,
    CASE WHEN u.owners > 1 THEN 'cross-owner'
      WHEN u.occurrences > 1 THEN 'same-owner-multiple'
      ELSE 'single-occurrence' END AS reuse_stratum
  FROM artifacts p JOIN repos r
    ON r.full_name = p.repo_full_name
  JOIN reuse u ON u.file_sha = p.file_sha
  WHERE p.dedup_primary = 1
    AND p.filename = 'SKILL.md' COLLATE BINARY
    AND p.frontmatter_valid = 1 AND p.content IS NOT NULL
    AND TRIM(p.content) <> ''
    AND r.metadata_fetched = 1 AND r.is_fork = 0
)
SELECT reuse_stratum, COUNT(*) AS groups,
  SUM(composition_fetched = 1 AND composition_truncated = 0)
    AS complete_folders
FROM frame GROUP BY reuse_stratum ORDER BY reuse_stratum;
```

Observed stratum	Content groups	Complete folders
cross-owner	2234	2040
same-owner-multiple	693	557
single-occurrence	8265	7482
Total	11,192	10,079

Sampling, annotation and provenance

Within each of the three strata, the runner sorts by SHA-256("cs599-label-v1:" + file_sha) and selects the first 80. This yields 240 rows with blank verification_level, evidence_lines, explicit_local_reference, resolution_status and notes fields. The form is a completed sampling artifact, not completed human annotation. The analyst must hide reuse context before coding.

The archive preserves the exact SQL, schema, observed results and manifest so the checkpoint evidence can be checked without trusting prose or manually transcribed counts. The raw sample is obtained from the pinned public source; the full release remains unqueried. A future study report must add completed labels, label validation, effect estimates and uncertainty, and peer feedback with a documented response.